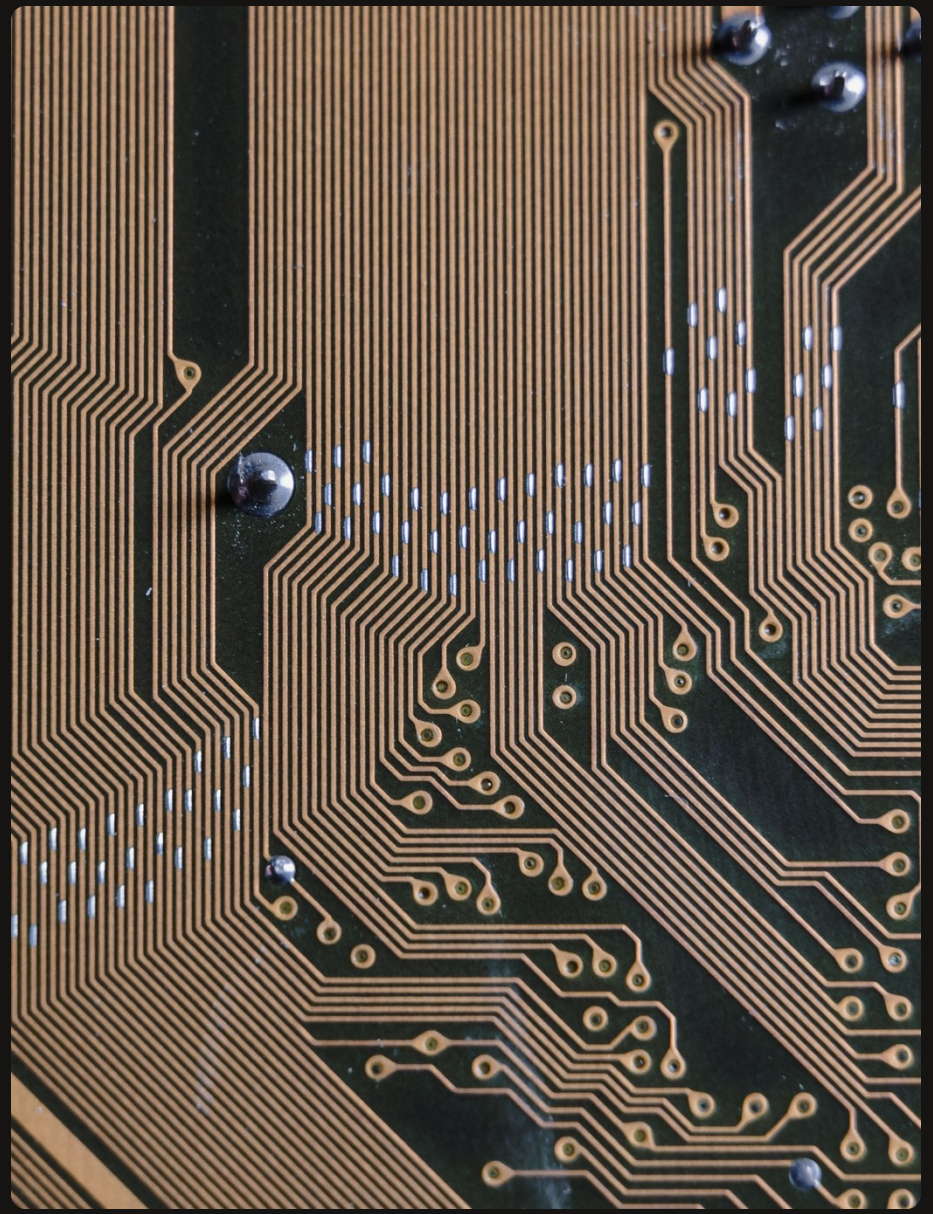
Tokito Companion

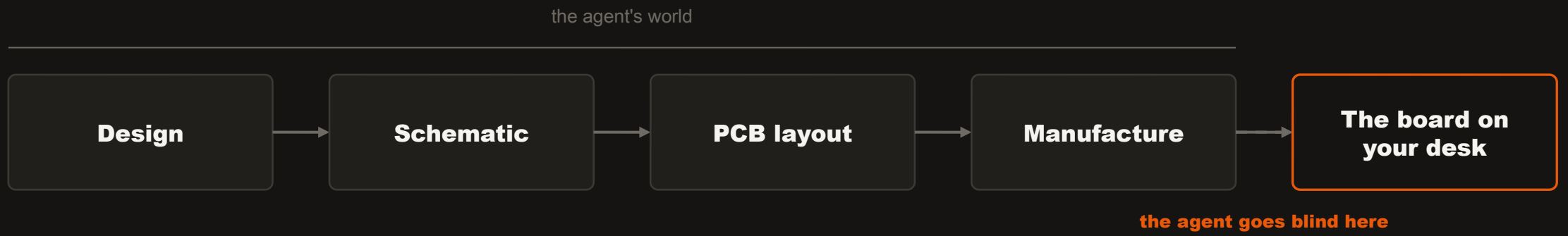
The physical intelligence layer for electronics engineering.

Built on Tokito: our AI-native electronics design environment.



Design is digital. Debugging is physical.

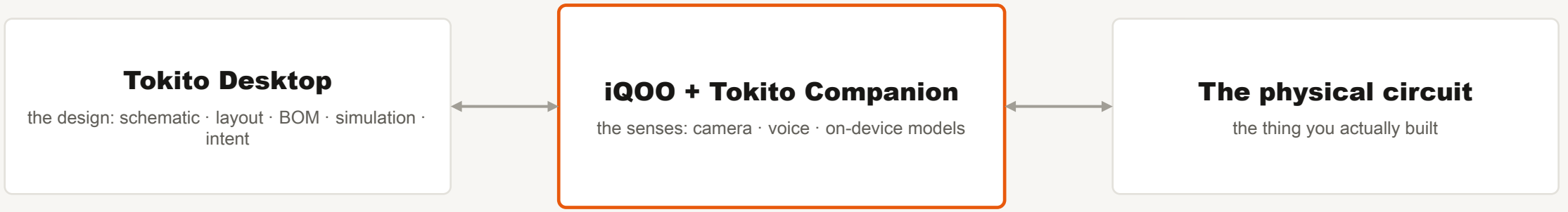
Tokito's agent holds the entire design: schematic, layout, BOM, simulation, intent. The moment that design becomes a physical board on the desk, the agent loses sight of it.



So debugging falls back to the old loop:

- multimeter
- datasheets
- manual inspection
- handwritten notes
- trial and error

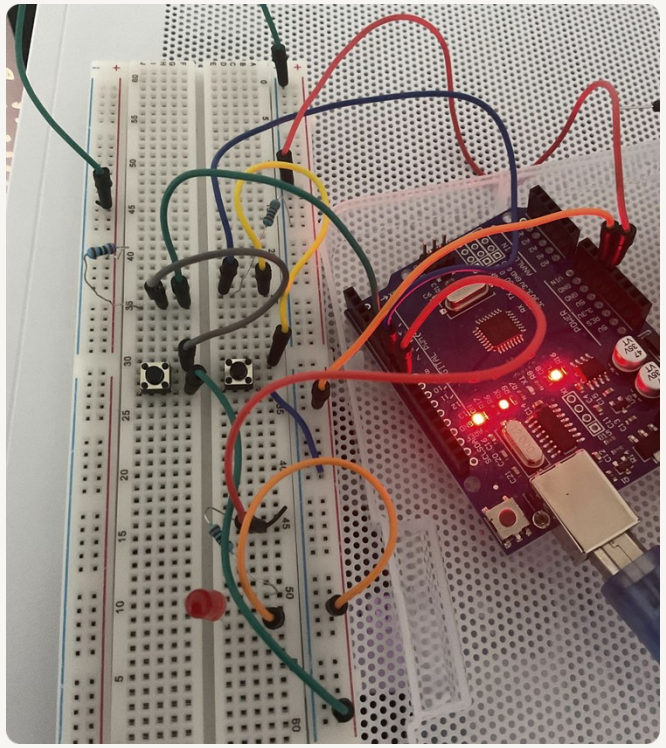
What if the engineering agent could see what you built?



Tokito Companion connects Tokito's desktop agent to the physical circuit on your desk, through the phone.

The phone is not a remote display, it is the agent's perception hardware.

Point the phone at the board. Ask what's wrong.



“Tokito, why isn't this LED turning on?”

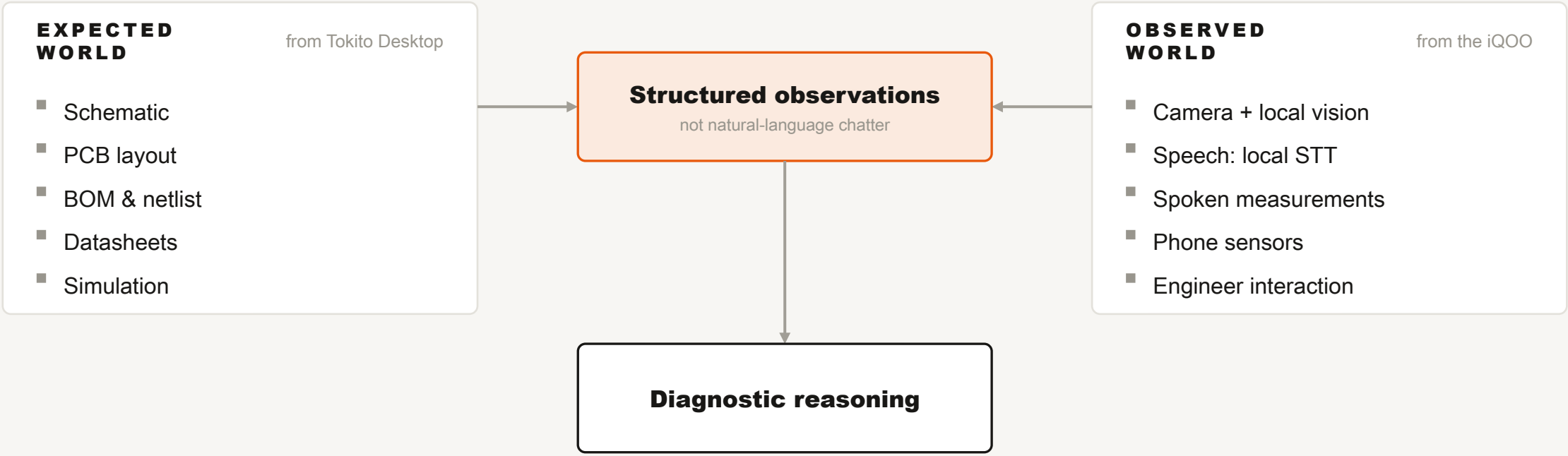
SEE	The camera maps the physical circuit against the design. **The LED is off.**
REASON	Tokito orders hypotheses from the schematic, check the supply first.
GUIDE	“Measure across these two points.” The phone highlights them on screen.
MEASURE	“4.96 volts.” `voltage · TP1 → GND · 4.96 V · source: voice`
VERIFY	Engineer fixes the fault. The camera confirms: **LED active** circuit matches the design.

A DEBUG SESSION SAMPLE

```
14:31:04  VIN = 5.08 V        14:31:31  3V3 = 0.31 V
14:32:02  3V3 → GND = 4.7 Ω     14:32:15  U2 reported hot
```

Tokito knows what should exist. The phone shows what does.

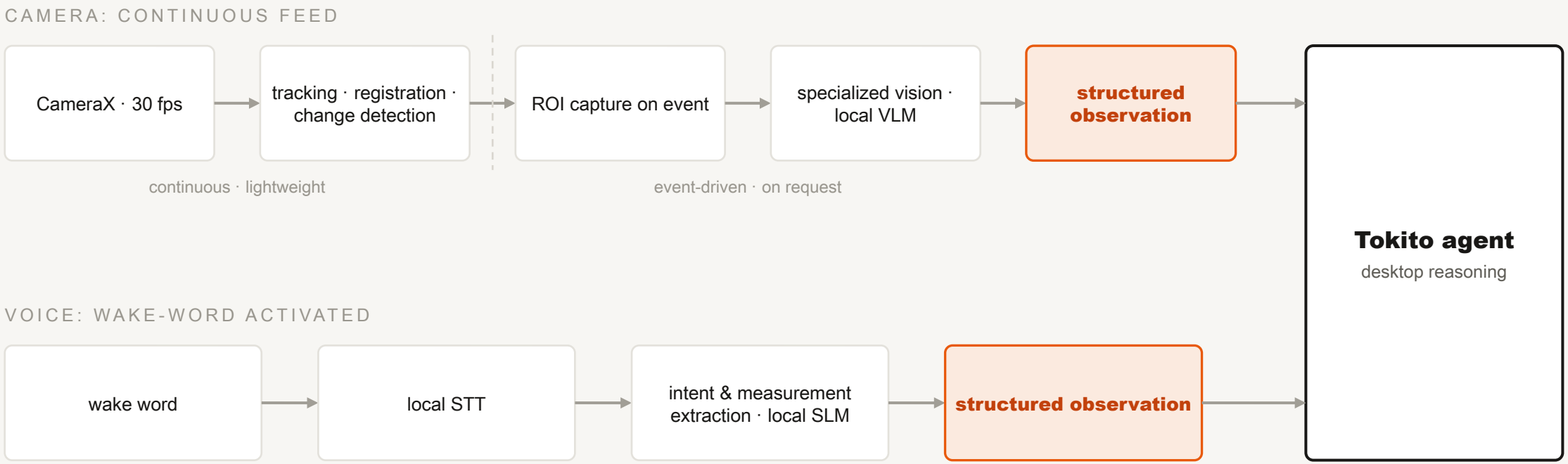
Debugging is a comparison problem, and both sides can be structured.



`design graph + observation graph + datasheet context + debug history → diagnostic reasoning`

A deliberate pipeline: not one big VLM.

Lightweight CV runs on the continuous feed. Heavy inference is event-driven.



This is what keeps latency, battery and thermal load sane on a phone.

The phone exposes the physical world as tools.

Tokito's agent calls the bench the way it calls any other tool.

locate_component()	find C14 on the board; return physical coordinates
inspect_component()	crop the region; check package, orientation, presence
read_led_state()	detect on / off and blink rate
request_measurement()	ask the engineer; parse the spoken value
verify_assembly()	compare placed parts against the BOM
verify_change()	watch a region until it changes, or doesn't

TOKITO DESKTOP → COMPANION

```
inspect_component({
  ref: “U7”,
  package: “LQFP48”,
  orientation: 90,
  near: [“C21”, “R14”, “C19”]
})
```

→ the phone crops U7's region and evaluates only that region.

Only the requested region is captured and evaluated, not the whole board, not every frame.

The board and the design are one object.

Every physical part links back to its design, in both directions.

DIGITAL → PHYSICAL

Select C14 in Tokito Desktop → (select → highlight) → Camera highlights C14 on the board

agent: “measure here” → test point lit

physical component ↔ PCB footprint ↔ schematic symbol ↔ net ↔ datasheet ↔ agent context

PHYSICAL → DIGITAL

Point at a part through the camera → (tap the board → open in the design) → Tokito opens symbol · footprint · datasheet · context

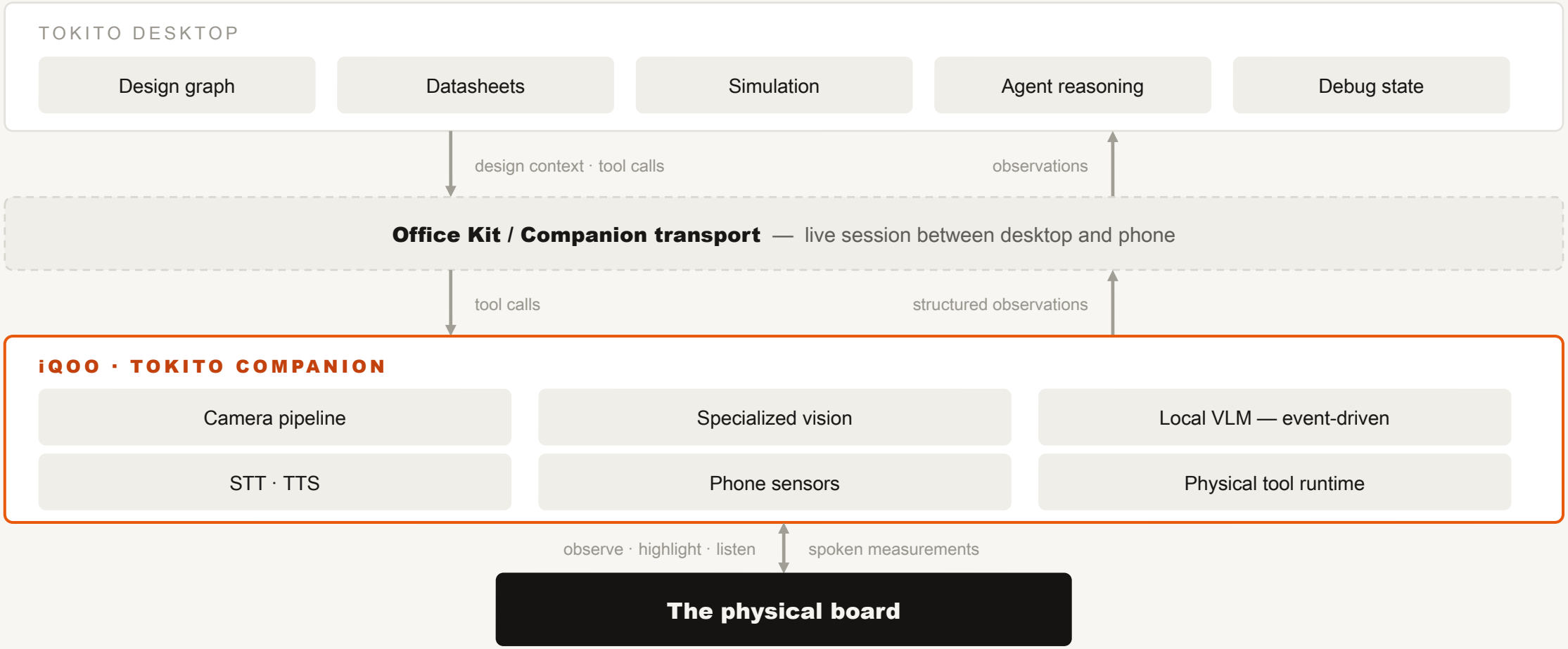
Generic object recognition finds “a resistor”. Tokito resolves which one, its net, its footprint, its datasheet.

Everything runs on the phone. On purpose.

- **Latency:** perception and speech stay in the loop, not a round-trip away.
- **Privacy:** unreleased designs and a continuous camera feed never leave the device.
- **Reliability:** the bench keeps working when the network doesn't.
- **Efficiency:** event-driven heavy inference keeps battery and thermals in check.
- **Local, open models:** the whole stack runs on-device, NPU-accelerated where available.

The full system.

Desktop reasoning, phone senses, the physical board, one closed loop.



What existed, what we will build, what comes next.

BEFORE THE HACKATHON

- Tokito: AI-native electronics design environment
- Companion: basic phone ↔ desktop streaming link

AT THIS HACKATHON

- Physical observation pipeline
- On-device vision + event-driven local VLM
- Voice measurement interface: STT · TTS
- Physical ↔ digital component mapping
- Structured observation graph
- Agent tool runtime for the bench

DESIGNED FOR NEXT

- Assembly inspection at bring-up
- Physical ↔ digital sync: expected 330 Ω, board shows 1 kΩ → “update the design?”
- USB-C & BLE instruments: multimeter, logic analyzer, scope, serial
- Automated measurement loops

TOKITO COMPANION

We gave the engineering agent eyes, ears, and a voice.

Engineering agents understand designs.

Now they understand what you built.

